An Improvement of the Method for Removing Erroneous Translation Rules in GA-ILMT

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Abstract

The goal of our research is to realize a machine translation system which is capable of adapting itself to the environment. For this purpose, we proposed a method of Machine Translation using Inductive Learning with Genetic Algorithms(GA-ILMT). Through the results of experiments, we confirmed the effectiveness of GA-ILMT. However, the system based on this method produces many erroneous translation rules that cannot be completely removed from the dictionary. Therefore, we need to improve how to remove the erroneous translation rules from the dictionary. For this reason, we improved the selection process by utilizing only the given translation examples. In this paper, we describe a improvement of the method for removing the erroneous translation rules in GA-ILMT and the results of evaluation experiments.

1 Introduction

Recently, a lot of research has been carried out towards the realization of a practical and high-quality machine translation systems. However, those are not practical enough. The main method used in machine translation systems is the Rule-based method [1]. However, there are several problems in the Rule-based method. The Rule-based method can not deal adequately with various linguistic phenomena due to the use of limited rules. Moreover, it also has difficulty in dealing with unregistered words due to the use of a non-dynamic dictionary. To resolve these problems, the Example-based method has been proposed in recent years [2, 3, 4, 5]. This method can improve translation ability by adding new translation examples. However, the Example-based method requires many translation examples to achieve a practical and high-quality translation.

The goal of our research is to realize algorithms with the same capability of language and knowledge acquisition as human beings [6, 7]. Towards this goal, we previously proposed a method of Machine Translation using Inductive Learning with Genetic Algorithms, and evaluated it [8]. In this method, translation rules are inductively acquired from only translation examples, and translation results are produced by using these rules. Therefore, this method has the advantage of the Example-based method which can improve translation ability by adding new translation examples. Also, many translation examples are automatically produced from only a small number of translation examples by crossover and mutation, which is the basic process of genetic algorithms. Therefore, we consider that our proposed method can solve the problem that the Example-based method requires many translation examples to do. Thus, our proposed method does not need to prepare grammatical rules as an initial condition. This method has ability to automatically gain knowledge by the learning method, and can improve the ability by applying genetic algorithms. We call this method GA-ILMT. We have already confirmed the effectiveness of GA-ILMT.

However, GA-ILMT has some problems. This main problem is that many erroneous translation rules are produced by applying genetic algorithms, and these rules cannot be immediately removed from the dictionary. This leads to many erroneous translation results. Therefore, we need to improve how to remove the erroneous translation rules. In this paper, we describe an improvement of the selection process which removes the erroneous translation rules in GA-ILMT, and the effectiveness of the improvement.

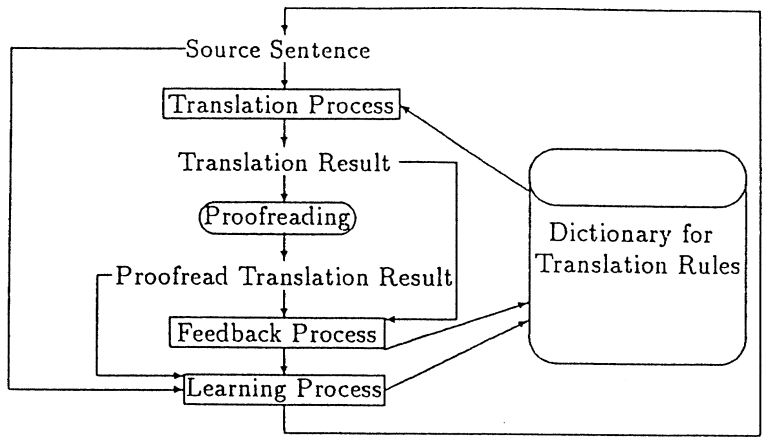


Figure 1: Outline of GA-ILMT

2 Outline of GA-ILMT

GA-ILMT can translate many different languages by merely changing to language of the translation examples, because this method aims to require translation rules from only character strings of translation examples. Figure 1 shows the outline of GA-ILMT.

First, in the case of English-Japanese, the user inputs a source sentence in English. Second, in the translation process, the system produces several candidates of translation results using translation rules extracted in the learning process. Third, the user proofreads the translated sentences if they include any errors. Fourth, in the feedback process, the system determines the fitness value of the translation rules used in the translation process and performs the selection process for erroneous translation rules. In the learning process, new translation examples are automatically produced by crossover and mutation, and various translation rules are extracted from the translation examples by inductive learning. This process uses one-point crossover and two-point crossover as the method of crossover. Repetition of the above-mentioned processes corresponds to generational replacement of the whole system, and the system continuously evolves to higher quality translation.

3 Translation Rule

In the learning process, new translation examples are automatically produced by crossover and mutation, and many translation rules are extracted from the translation examples. In crossover, two translation examples which have common parts are selected, and crossover positions are the common parts. Crossover uses each common part of the English and the Japanese sentences of translation examples. Figure 2 shows examples of one-point crossover. Italic means pronunciation of Japanese. In Figure 2, "likes" is the common part in the two English sentences, and "*wa*" and "*ga suki desu*" are the common parts in the two Japanese sentences. Therefore, "likes" and "*wa*" are the crossover positions. And new translation examples are produced by one-point crossover. Next, one-point crossover are performed for "like" and "*ga suki desu*". However, the translation examples which are produced have the same character strings as the source sentences, and therefore, these translation examples are not inputted into the dictionary.

(1)Selection from translation examples
ENGLISH JAPANESE
(He likes tennis.
:*Kare wa tenisu ga suki desu.*)
(She likes tea.
:*Kanojyo wa ocha ga suki desu.*)
(2)The crossover of the English sentences
He likes / tennis. → He likes tea.
She likes \ tea. She likes tennis.
(3)The crossover of the Japanese sentences
Kare wa / *tenisu ga suki desu.*
Kanojyo wa \ *ocha ga suki desu.*
Kare wa ocha ga suki desu.
→*Kanojyo wa tenisu ga suki desu.*
(4)The translation examples produced
ENGLISH JAPANESE
(He likes tea.
:*Kare wa ocha ga suki desu.*)
(She likes tennis.
:*Kanojyo wa tenisu ga suki desu.*)

Figure 2: Examples of one-point crossover

And, the system extracts the common and different parts from the character strings of all translation examples. These common and different parts are used as translation examples. Figure 3 shows the examples of the extracted translation rules. In Figure 3, these translation rules are extracted by using

new translation examples. The extracted translation rules are sorted out based on the number of variables, and registered in the dictionary. Also, the produced translation examples are registered in the dictionary as translation rules after being learnt.

(1)The translation rules which do not include variable

ENGLISH	JAPANESE
(She likes tennis.	;*Kanojyo wa tenisu ga suki desu.*)
(He likes tennis.	;*Kare wa tenisu ga suki desu.*)
(She likes tea.	;*Kanojyo wa ocha ga suki desu.*)
(He likes tea.	;*Kare wa ocha ga suki desu.*)
(He	;*kare*)
(She	;*kanojyo*)
(tea	;*ocha*)
(tennis	;*tenisu*)

(2)The translation rules whcih include one variable

ENGLISH	JAPANESE
(He likes @0.	;*Kare wa @0 ga suki desu.*)
(She likes @0.	;*Kanojyo wa @0 ga suki desu.*)
(@0 likes tea.	;*@0 wa ocha ga suki desu.*)
(@0 likes tennis.	;*@0 wa tenisu ga suki desu.*)

(3)The translation rules which include two variables

ENGLISH	JAPANESE
(@0 likes @1.	;*@0 wa @1 ga suki desu.*)

Figure 3: Examples of extracted translation rules

4 Improvement of the Method for Removing Erroneous Translation Rules

4.1 Previous Selection Process

The selection process is performed in the feedback process described in Section 2. First, the system evaluates the translation results using the translated sentences which have been proofread. As a result, the system adds one to the correct translation frequency when the translation results have the same character strings as the proofread translation results, and adds one to the erroneous translation frequency when the translation results have different character strings to the proofread translation results. Second, the system determines the fitness value of the translation rules used in translation using these correct and erroneous translation frequencies. The fitness value is calculated by the fitness function as follows:

$$\text{Fitness value}(\%) = \frac{\text{The correct translation frequency}}{\text{The number of uses}} \times 100 \quad (1)$$

Third, the system removes the erroneous translation rules using the fitness value. The condition of the removal is that the number of uses is over 5 and the fitness value is under 25%. These thresholds were determined by preliminary experiments.

4.2 Problem

The previous selection process has some problems. The main problem is that many erroneous translation rules exist in the dictionary. The translation rules are evaluated only when they are used in the translation process, and only a minority of the whole translation rules in the dictionary are used. The number of uses of many translation rules is under 5. Therefore, the removed erroneous translation rules are a minority of all the erroneous translation rules in the dictionary. Also, most of the removed erroneous translation rules consist of a few words. Most of the erroneous translation rules which consist of many words have different character strings or structure to the source sentence, and cannot be selected as the translation rules which apply to the source sentence in the translation process. Therefore, the system based on the previous selection process has difficulty in removing the erroneous translation rules which consist of many words from the dictionary. Figure 4 shows the examples of erroneous translation rules which are produced by one-point crossover.

(I don't like volleyball.
; *Watashi wa basukettobooru ga suki desu.*)
(You like basketball.
; *Anata wa bareibooru ga suki dewa nai.*)

Figure 4: Examples of erroneous translation rules

In Figure 4, the first example is the translation rule where the English sentence is the negative sentence and the Japanese sentence is the affirmative sentence (In Japanese "I like basketball."), and the other example is the translation rule where the English sentence is the affirmative sentence and the Japanese is the negative sentence (In Japanese "You don't like volleyball.") The reason for this error is that the system does not know the difference of the position between Japanese and English negative phrases. The human being can judge whether these translation rules are the erroneous translation rules or not by using grammatical knowledge of language. Therefore, we improve the selection process to immediately remove the erroneous translation rules.

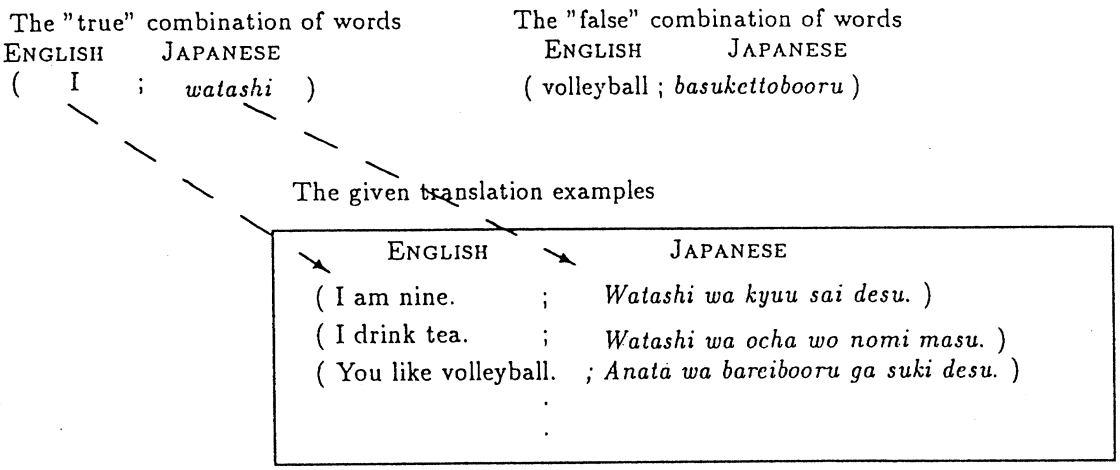


Figure 5: Examples of combinations of words

4.3 Improvement

In genetic algorithms [9, 10], the execution of the selection process depends on whether an individual adapts to the environment or not. The individual which does not adapt to the environment dies out in the selection process. In GA-ILMT, the given translation examples correspond to the environments. Therefore, our proposed method utilizes only given translation examples. Also, this method does not require any analytical knowledge as an initial condition, such as an English-Japanese dictionary. The reason is that the method which uses analytical knowledge has the problem of robustness, such as difficulty in dealing with unregistered words. Therefore, we consider that our proposed method can remove most of the erroneous translation rules by the learning method which utilizes only given translation examples.

Our proposed improvement evaluates the translation rules by utilizing only the given translation examples directly. Namely, the system determines the erroneous words used in translation which are included in all combinations of the English and Japanese words in a translation rule. The combination of words may produce an error when it does not exist in all given translation. Figure 5 shows examples of the combinations of words.

In Figure 5, the combination of words which are "I" and "*watashi*" are possible to be correct when it exists in a given translation example. The "*watashi*" corresponds to "I". On the other hand, the combination of words which are "volleyball" and "*basukettobooru*" are possible to be erroneous when it does not exist in all given translation examples. The "*basukettobooru*" corresponds to "basketball".

4.4 Description of the Improvement Method

The detail of the improvement was described in Subsection 4.3 is as follows:

1. The system determines whether the translation rule is error or not.

(a) The system produces all combinations of words between the English and Japanese words in a translation rule.

(b) The system determines whether the combinations of words exist in given translation examples or not.

(c) The system gives "existence" for the combinations of words when they exist in a given translation example, and gives "non-existence" for the combinations of words when they do not exist in all given translation examples.

In "non-existence" combinations, there are two possibilities for faults as follows : (1) an erroneous word used in translation (2) an erroneous combination. For these faults, the system performs the process (d).

(d) The system determines the erroneous combinations and deletes them. The detail of this process is as follows:

1) The system changes the combinations of words between the English and the Japanese words in all "non-existence" combinations.

2) The system searches the changed combinations of words in translation rules in the dictionary.

3) The system changes "non-existence" for "existence" in the results of evaluations for the "non-existence" combinations of words when the changed combinations of words exist in the translation rules in the dictionary.

(e) The system changes "true" for "existence"

Table 1: Example of the erroneous translation rule "she is not my teammate.; *kanojyo wa boku no chiimumeito desu.*"

word	she	is	not	my	teammate
kanojyo	existence	existence	existence	existence	non-existence
wa	existence	existence	existence	existence	existence
boku no	existence	existence	existence	existence	existence
chiimumeito	non-existence	existence	existence	existence	existence
desu	existence	existence	non-existence	existence	existence

based on heuristics which the combinations of words are possible to be correct when they exist in a given translation example. And the system changes "false" for "non-existence" based on heuristics which the combinations of words are possible to be erroneous when they does not exist in all given translation examples.

(f) The system determines the rate of error based on the number of the "false" combinations of words. The rate of error is calculated as follows:

$$\text{Rate of error}(\%) = \frac{\text{The number of "false" combinations of words}}{\text{The number of combinations of words}} \times 100 \quad (2)$$

2. The system removes the erroneous translation rule from the dictionary.

The system determines the translation rules for which the rate of error is over the threshold, and removes them from the dictionary as the erroneous translation rule. The condition of removal is defined as follows:

$$\text{Rate of error} \geq x \qquad (x : \text{Threshold}) \qquad (3)$$

Next, we describe an example of selection process. In the erroneous translation rule "she is not my teammate.; *kanojyo wa boku no chiimumeito desu.*" English sentence is the negative sentence and Japanese sentence is the affirmative sentence (In Japanese "she is my teammate.") First, the system determines whether the combinations of words exist in given translation examples or not. Table 1 shows the result.

In Table 1, there are four "non-existence" combinations. The combination of words which are "she" and "*chiimumeito*" and the combination of words which are "teammate" and "*kanojyo*" are erroneous combinations. The "*chiimumeito*" corresponds to "teammate", and the "*kanojyo*" corresponds to "she". Therefore, the system determines their erroneous combinations and deletes them. For this propose, the system changes the combinations of words between the English and the Japanese words in all "non-existence" combinations. Figure 6 shows the examples of the changed combinations. And the system searches the changed combinations of words in translation rules in the dictionary. As a result, the changed combination of words which are "she" and "*kanojyo*" and the changed combination of words which are "teammate" and "*chiimumeito*" existed in the translation rules in the dictionary. Therefore, the system changed "non-existence" for "existence" in the results of evaluations for them. Second, The system changes "true" for "existence" based on heuristics which the combinations of words are possible to be correct when they exist in a given translation example. And the system changes "false" for "non-existence" based on heuristics which the combinations of words are possible to be erroneous when they does not exist in all given translation examples. Table 2 shows the result of a final evaluation for the erroneous translation rule.

Table 2: Example of the result of a final evaluation for the erroneous translation rule "she is not my teammate.; *kanojyo wa boku no chiimumeito desu.*"

word	she	is	not	my	teammate
kanojyo	true	true	true	true	true
wa	true	true	true	true	true
boku no	true	true	true	true	true
chiimumeito	true	true	true	true	true
desu	true	true	false	true	true

Third, the system removes the erroneous translation rule from the dictionary. In this erroneous translation rule, the rate of error is 4.0% (1/25×100). The translation rule is removed from the dictionary if the threshold is over 4.0%.

Thus, this method of improvement evaluates the translation rules by utilizing only the given translation examples directly without requirement of analytical knowledge as an initial condition, and immediately removes the erroneous translation rules. There-

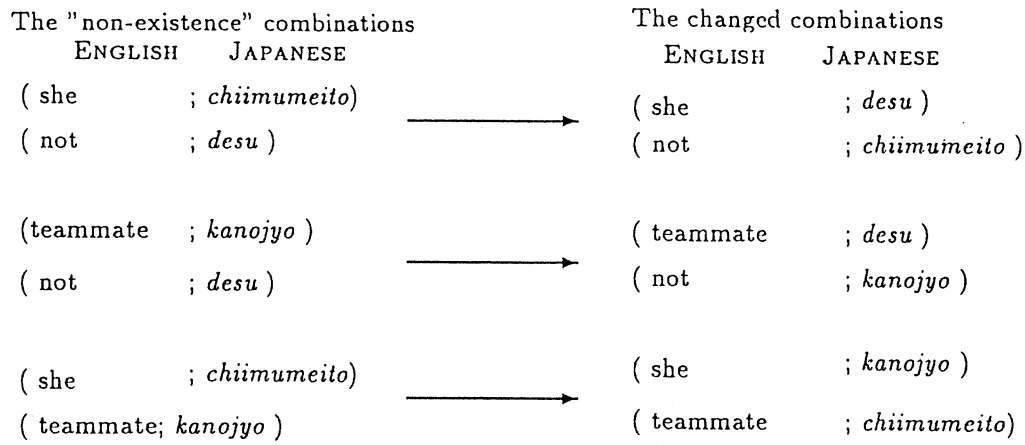


Figure 6: Examples of the changed combinations

fore, the purpose of this method is not the predication the correct words used in translation. Also, although this method assumes the correct words used in translation by using translation rules in the dictionary (e.g. "she" and "*kanojyo*", "teammate" and "*chiimumeito*"), the purpose of it is that the system determines the erroneous translation combinations among "non-existence" combinations.

5 Experiments

We evaluate the effectiveness of our proposed improvement for removing the erroneous translation rules in the selection process of GA-ILMT.

5.1 Standards of Evaluation

We describe the standards of evaluation for the translation rules. The erroneous translation rules has three kinds as follows:

1. The translation rule has the erroneous words used in translation between the English and the Japanese word.

2. The translation rule has the erroneous words used in translation which are the lack of the English words or the Japanese words.

3. The translation rule has both 1 and 2.

Figure 7 shows the example of evaluation for the translation rules.

In Figure 7, the first example is the translation rule where "*kare*" in Japanese corresponds to "he" in English. The second example is the translation rule where "*futtobooru*" in Japanese corresponds to "football" in English. The third example is the translation

(1) The errneous word used in translation
(volleyball ; *basuketbooru*)
(2) The lack of the English word or Japanese word
(like football ; *futtobooru*)
(3) The both 1 and 2
(do not like volleyball ; *yakyuu ga suki*)

Figure 7: Examples of evaluation for the translation rules

rule where "*yakyuu ga suki*" in Japanese corresponds to "like baseball" in English.

The goal of our proposed method is that the system immediately removes the erroneous translation rules that the human can judge. Therefore, article (e.g. "a", "the") in English and postpositional article (e.g. "*wa*", "*o*") in Japanese do not evaluate because there is usually no corresponding word as word used in translation.

5.2 Method of Experiments

In the experiments, 1,010 translation examples were used as data. Of these, 500 were used to decide the best threshold. The examples were taken from a textbook [11] for first-grade junior high school students. All of the translation examples were processed by the method outlined in Figure 1. The initial dictionary was empty. The experiments were carried out with and without the improvement for the selection pro-

Table 3: Precision and recall which were used some threshold

Threshold	Precision	Recall
2%	92.6%	66.4%
5%	94.5%	61.3%
10%	96.7%	52.0%
15%	97.6%	40.5%

Table 4: Precision and recall with and without the improvement for the selection process

Kind	Precision	Recall
Method without improvement	88.1%	1.5%
Method with improvement	97.0%	63.9%

cess described in Section 4.3. Also, the human judged whether the removed translation rules were the erroneous translation rules or the correct translation rules.

5.3 Results of Experiments

Table 3 shows the precision and the recall which were gotten by using some thresholds. The precision is the percentage of removed erroneous translation rules to removed translation rules, and the recall is the percentage of removed erroneous translation rules to erroneous translation rules in the dictionary. Table 4 shows the precision and the recall with and without the improvement for the selection process, and Table 5 shows the rate of translation with and without the improvement for the selection process. In Table 4 and Table 5, 5% is used as the threshold in the results of the method with the improvement for the selection process.

Table 5: Rate of translation

Kind	Effective translation	Ineffective translation
Method without improvement	39.5%	60.5%
Method with improvement	41.2%	58.8%

5.4 Discussion

In Table 3, the precision increases by about 2%. On the other hand, the recall decreases by about 10% when the threshold is over 5%. However, the recall decreases by 5.1% when the threshold is under 5%. Therefore, we decided that the best threshold was 5% because the rate of the decrease in the recall is lower when the threshold is under 5% than over 5%.

In Table 4, the precision increases from 88.1% to 95.7% and the recall increases from 1.5% to 62.8%. These results show that our proposed improvement could resolve the problem that the removed erroneous translation rules are a minority of all the erroneous translation rules in the dictionary. In the previous selection process, the translation rules are evaluated only when they are used in the translation process. Therefore, the translation rules which are not used in the translation process are never removed from the dictionary. However, the proposed improvement can evaluate all of the produced translation rules by utilizing only the given translation examples. In the previous selection process, the rate of the evaluated translation rules is 1.6% of the translation rules in the dictionary.

Also, our proposed improvement could resolve the problem that most of the erroneous translation rules which consist of many words cannot be removed from the dictionary. In the previous selection process, the average number of words in a removed translation rule is 4.6. However, the average number of words in a removed translation rule is 8.0 in our proposed improvement. The average number of words in a translation rule in the dictionary is 7.9. Therefore, we consider that our proposed improvement is effective for the erroneous translation rules which consist of many words.

In Table 5, the rate of translation increases from 39.5% to 41.2%. In GA-ILMT, the system ranks from the first to the tenth of the candidates for the translation results. The effective translation results are included in these ranks. In the effective translation of method without improvement, the number of the translation results that the rank was under the first was 185. Of these, 114 improved their ranks in the translation results. Therefore, the rate of the translation results that the rank improved was 61.6%. The reason for this is that the system based on our proposed improvement can remove many erroneous translation rules in the dictionary.

And, the erroneous translation results decreases by 64.9%.

5.5 Conclusion

In this paper, we proposed an improvement in the selection process of GA-ILMT. The results of evalua-

tion experiments showed that the precision increases from 88.1% to 97.0% and the recall increases from 1.5% to 63.9%. Also, the erroneous translation results decreases by 64.9%. These results show that our proposed improvement method can immediately remove the erroneous translation rules, and does not require any analytical knowledge as an initial condition. Therefore, we confirmed that our proposed improvement is effective to improve the selection process of GA-ILMT. In the future, we plan to experiment using the various data, and improve GA-ILMT to get the high correct rate and quality of translation.

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